

Data & AI

DATA AND AI PRACTICE: CORE CAPABILITIES AND DELIVERY STANDARDS

PRACTICE PHILOSOPHY: "AI ONLY WORKS WHEN PLATFORMS, DATA, AND PEOPLE ARE READY"

Gauri's Data & AI practice is built on the foundational principle that artificial intelligence only creates measurable business value when an organisation's underlying platforms, data, and people are genuinely ready for it.

This contrasts with industry narrative of AI as transformative bolt-on technology. Gauri positions data engineering, governance, and organisational readiness as prerequisites to meaningful AI deployment, not afterthoughts.

The practice is structured around three complementary capability areas:

1. Data Engineering (foundations)
2. Data Analytics (insight and decision support)
3. Generative AI (intelligent automation and augmentation)

DATA ENGINEERING: GOVERNED FOUNDATIONS

STRATEGIC POSITIONING

Data engineering is the foundation of Gauri's AI readiness. Without reliable, governed data accessible to AI systems, generative and agentic AI produce hallucinations and poor decisions. Gauri emphasises data as a first-class product requiring the same governance discipline as production software.

LAKEHOUSE ARCHITECTURE: UNIFIED STORAGE AND GOVERNANCE

WHAT IS A LAKEHOUSE?

Lakehouse architecture combines the low-cost, flexible storage of a traditional data lake with the governance, schema enforcement, and query performance of a data warehouse. This hybrid approach offers:

Cost Advantages: Object storage (S3, Azure Blob, GCS) pricing vs. warehouse licensing

Flexibility: Schema-on-read capability for exploratory analytics

Governance: Schema-on-write constraints ensuring data quality

Performance: Optimised query engines (Parquet, Delta Lake, Apache Iceberg)

Scalability: Infinite horizontal scaling for petabyte-scale datasets

TYPICAL GAURI LAKEHOUSE ARCHITECTURE

Data Sources (SAP, Salesforce, third-party systems) → Cloud Object Store (Snowflake, Databricks, Hyperscaler native) → Governed Layer (Bronze/Silver/Gold tables) → Analytics and AI consumption

IMPLEMENTATION PATTERNS

Ingestion Layer (Bronze): Raw data from source systems ingested as-is

Transformation Layer (Silver): Data cleansing, deduplication, joining, and business logic application

Analytics Layer (Gold): Business-ready tables aligned to reporting and analytics needs

AI Layer: Curated datasets for model training and real-time scoring

CLOUD DATA PLATFORM SELECTION

Gauri's lakehouse architectures are built on major hyperscale cloud data platforms selected on a client-by-client basis:

- Snowflake: Mature ecosystem, extensive tooling, cloud-agnostic
- Databricks: Delta Lake foundation, strong ML integration, Apache Spark native
- Hyperscaler Native Solutions: Azure Synapse, Google BigQuery, AWS Redshift Spectrum (aligned with existing cloud commitments)

Selection Criteria: Existing cloud commitments, SAP and Salesforce integration capabilities, team skill base, total cost of ownership

SEMANTIC MODELS: BUSINESS-FRIENDLY DATA GOVERNANCE

THE PROBLEM THEY SOLVE

Large organisations frequently have conflicting definitions of core metrics. "Active customer," "revenue," and "order value" are calculated differently across SAP, Salesforce, and legacy systems, producing different answers in different reports. This ambiguity undermines trust in analytics and AI systems.

GAURI'S APPROACH

Semantic models (or "business intelligence models") define single, authoritative definitions of key metrics. They serve as the translation layer between raw data and business intelligence tools.

TYPICAL SEMANTIC MODEL CONTENT

Entities: Customer, Order, Product, Sales Territory, Time Period

Measures: Revenue, Unit Sales, Gross Margin, Customer Acquisition Cost, Customer Lifetime Value

Dimensions: Customer Segment, Product Category, Geography, Sales Channel, Fiscal Period

Hierarchies: Product → Category → Segment, Geography → Region → Territory

Calculations: Gross Margin = Revenue - Cost of Goods Sold, Customer LTV = Sum of Margins over 5 years

INTEGRATION WITH DATA PLATFORMS

Semantic models are typically implemented as:

- Semantic layer tools (Looker, Tableau semantic layer, ThoughtSpot modeling)
- dbt (data build tool) models with documentation and tests
- Dimensional data warehouse schemas (star schemas, snowflake schemas)

BUSINESS VALUE

Removes ambiguity and reporting delays caused by definition disagreements

Enables self-service analytics without centralised reporting team bottleneck

Provides input to AI systems ensuring they reason from consistent, auditable definitions

Speeds time-to-insight by pre-calculating common metrics

ROPER RHODES CASE STUDY APPLICATION

Bathroom products manufacturer and retailer defined semantic model for supply chain and sales metrics

Enabled self-service analytics for supply chain and inventory visibility

Integrated with ThoughtSpot for fact-based decision-making

DATA QUALITY, GOVERNANCE, AND LINEAGE

FIRST-CLASS DELIVERABLES, NOT AFTERTHOUGHTS

Gauri's data engineering work typically includes data quality rules, lineage tracking, and access governance as first-class deliverables. These are designed into pipelines from inception, not bolted on after production deployment.

DATA QUALITY FRAMEWORK

Quality Rules: Validity (value in allowed set), Completeness (not null), Consistency (matches other source), Accuracy (correct or within tolerance)

Monitoring: Automated daily quality checks with alerts when thresholds breach

Root Cause Analysis: Investigation and remediation process when quality fails

Data Quality Reporting: Business user visibility to data quality issues affecting their reports

Remediation Tracking: Documentation of quality issues and their resolution

LINEAGE TRACKING

Column-Level Lineage: Tracking transformation of each database column from source system through analytics layer

Impact Analysis: Identifying downstream reports affected by a data issue

Change Impact Assessment: Understanding consequences of modifying a calculation

Compliance Reporting: Demonstrating data traceability for regulatory audit

ROLE-BASED ACCESS GOVERNANCE

Data Classification: Classifying data by sensitivity (public, internal, confidential, restricted)

Role-Based Access: Users granted access based on job function, not individual negotiation

Attribute-Based Access: Filtering data visible to users based on their attributes (e.g., regional managers only see their region)

Audit Logging: Recording all access to sensitive data for compliance investigation

Dynamic Masking: Hiding sensitive values (SSN, credit card) from non-authorised users

SETU-AI: DOCUMENT INTELLIGENCE PRODUCT

WHAT IS SETU-AI?

Setu-AI is Gauri's flagship generative-AI-powered product that transforms "inbound documents from any source into clean, structured data." It applies large language model-based document understanding to unstructured inbound content and outputs structured records ready to load into SAP, Salesforce, or client data platforms.

TECHNICAL FOUNDATION

LLM-Based Document Understanding: Natural language processing models trained on business document examples

Optical Character Recognition (OCR): Processing scanned and handwritten documents

Document Structure Recognition: Identifying tables, fields, and document sections

Entity Extraction: Identifying and extracting key fields (invoice number, amount, date, supplier)

Validation and Correction: Comparing extracted values against known reference data

DOCUMENT TYPES SUPPORTED

Invoices (supplier invoices for payables automation)

Purchase Orders (customer POs for order entry)

Contracts and Agreements (for compliance and obligation tracking)

Forms (application forms, registration documents)

Delivery Notes (for goods receipt matching)

Expense Receipts (for expense reimbursement)

Loan Documents (mortgage, personal loan documentation)

Custom Business Forms (client-specific document types)

REPRESENTATIVE USE CASES

Accounts Payable Automation: Automatically extract invoice data (invoice number, amount, date, supplier, line items) and post to SAP

Order Entry Acceleration: Extract customer PO details and create sales orders in Salesforce or SAP

Contract Obligation Tracking: Extract key terms (renewal dates, price adjustments, delivery dates) for contract lifecycle management

Student Application Processing: Extract application data from forms and transcripts into education CRM

Healthcare Compliance: Extract lot and serial numbers from supplier documentation for traceability

BUSINESS IMPACT METRICS

Medivance Surgical Supplies Case Study: Purchase-order-to-delivery-note matching time cut by more than half

Wentworth Regional College Case Study: 80% reduction in manual re-keying of applicant data during peak enrolment

Alderton Precision Components Case Study: 70% reduction in manual invoice data entry, freeing finance team for exception handling

INTEGRATION WITH SAP AND SALESFORCE

Setu-AI is deployed alongside SAP Business One, S/4HANA, and Salesforce implementations to automate document intake

API-based integration: Documents automatically flow to SAP or Salesforce as structured records

Validation loops: Extracted data validated against SAP reference data (supplier master, product master) before posting

Exception handling: Extraction confidence scores below threshold route to human review

Continuous learning: Extracted data reviewed and corrected by users, feedback used to improve model

DATA ANALYTICS: DECISION SUPPORT AND INSIGHTS

ANALYTICS CAPABILITY POSITIONING

Gauri's data analytics capability turns governed data foundations built by data engineering into decision-support tools: dashboards, self-service exploration, and fact-based reporting that replace spreadsheet-driven or gut-feel decision-making.

FROM DESCRIPTIVE TO PREDICTIVE ANALYTICS

DESCRIPTIVE ANALYTICS: WHAT HAPPENED?

Historical reporting and dashboards answering "what were results this month," "which products sold best," "which territories exceeded quota"

Typical Deliverables:

- Sales dashboards (revenue by territory, product, customer segment)
- Operational dashboards (manufacturing yield, inventory turns, delivery performance)
- Financial dashboards (P&L; by business unit, cash position, headcount forecast)
- Supply chain dashboards (inventory levels, fulfillment rates, supplier performance)

PREDICTIVE ANALYTICS: WHAT WILL HAPPEN?

ML models forecasting future outcomes: demand forecasting, churn risk, lead scoring, pipeline forecasting

Typical Applications:

- Demand Forecasting: Predicting future sales volume by product, geography, time period
- Sales Pipeline Forecasting: Predicting quarterly revenue based on weighted opportunities
- Customer Churn Risk: Identifying at-risk customers for proactive retention
- Inventory Demand: Predicting inventory requirements to optimise stock levels

PRESCRIPTIVE ANALYTICS: WHAT SHOULD WE DO?

Recommendations on optimal actions given predictions and constraints (resource optimisation, pricing, routing)

Typical Applications:

- Dynamic Pricing: Optimising prices based on demand forecasts and competitor pricing
- Route Optimisation: Determining optimal technician routes minimizing travel time
- Inventory Allocation: Determining optimal inventory distribution across warehouses
- Workforce Scheduling: Recommending optimal shift assignments based on demand forecasts

ROPER RHODES CASE STUDY

ThoughtSpot search-driven analytics platform combined with Gauri data engineering and SAP integration

Bathroom-products manufacturer and retailer gained real-time, self-service visibility over supply chain and sales

Integration with Slim4 supply-chain tool for visibility across fulfillment

ANALYTICS ENGAGEMENT DELIVERY

1. Metrics Definition: Identifying key metrics that matter to client's leadership team (often tied to semantic model)
2. Tool Selection: Implementing BI or search-driven analytics tool suited to client technical maturity
3. Dashboard Design: Building role-specific dashboards (executive, manager, individual contributor)
4. Training: Training business users to self-serve rather than depend on central reporting team
5. Continuous Improvement: Iterative refinement based on user feedback and changing business needs

SECTOR-SPECIFIC ANALYTICS PRACTICE

Manufacturing: Supply chain visibility, production yield analysis, inventory optimisation

Financial Services: Risk and portfolio reporting, regulatory compliance reporting, profitability analysis

Retail: Channel performance (online, store, trade), customer lifetime value, merchandise performance

Healthcare: Patient outcomes, resource utilisation, cost per procedure

GENERATIVE AI: INTELLIGENT AUTOMATION AND AUGMENTATION

GENERATIVE AI POSITIONING

Generative AI sits at the top of Gauri's "AI-ready enterprise" stack. It is deployed only after platforms are modernised and data is governed, ensuring AI systems have high-quality inputs to reason from.

Gauri positions itself against ungoverned, speculative AI experimentation. Board-level framing: "How do we use AI without betting the business on it?"

GAURI'S GENERATIVE AI STRATEGY

Rather than building generative AI from scratch for every client, Gauri primarily implements and customises the generative AI capability already embedded in the platforms it specialises in:

- Salesforce Einstein (email and case-summary generation, sales insights)
- SAP Business AI (embedded copilots across SAP's cloud ERP suite)
- Setu-AI (document intelligence and extraction)

This approach leverages platform vendor investment and avoids reinventing capabilities that are better done at platform level.

REPRESENTATIVE USE CASES

1. Document Intake and Extraction (Setu-AI)

- Invoices and purchase orders to structured data
- Customer forms to customer records
- Contracts to obligation tracking

2. Salesforce Einstein Use Cases

- AI-assisted case and email drafting inside Salesforce Service Cloud
- AI-summarised sales and pipeline insights inside Sales Cloud
- Lead scoring and recommendation

3. SAP Business AI Use Cases

- Copilot-assisted process guidance inside SAP S/4HANA Cloud
- Procurement assistant for purchase order creation
- Demand forecasting assistance
- Expense and invoice processing assistance

GOVERNANCE-FIRST IMPLEMENTATION

Model Risk Assessment: Evaluating generative AI models for bias, hallucination risk, and data privacy

Use Case Validation: Validating proposed AI use cases against business value and risk criteria (Gauri internal AI review board)

Data Governance: Ensuring AI systems access only governed, high-quality data

Audit Trails: Recording all AI-assisted decisions for compliance investigation

Human-in-the-Loop: Maintaining human oversight of AI decisions, especially for high-consequence actions

Testing and Validation: Validating AI model accuracy before production deployment

Continuous Monitoring: Tracking model performance in production and retraining when accuracy drifts

CLOUD SERVICES AND INFRASTRUCTURE

CLOUD PLATFORMS FOR DATA ENGINEERING

Gauri's lakehouse-based data architecture is built on cloud object storage and compute:

- Hyperscale Cloud Providers: AWS, Azure, Google Cloud selected based on client preference
- Managed Services: Snowflake, Databricks, Fabric (managed platforms abstracting infrastructure)
- Serverless Compute: Spark, Presto, Trino for query execution without server management

CLOUD DATA INTEGRATION

Integration tooling connecting cloud data platforms to SAP and Salesforce sources:

- Talend: ETL platform with SAP and Salesforce connectors
- MuleSoft: API-based integration for real-time data sync
- SAP Landscape Transformation: Native SAP data replication tool
- Salesforce Connector: Native cloud connectors with Salesforce

MIGRATION AND MODERNISATION PATTERNS

For organisations moving off legacy on-premise data infrastructure:

- Infrastructure assessment and sizing
- Data migration planning and validation
- Cutover and hypercare support

- Performance tuning and optimisation
- Ongoing managed services

DATA AND AI PRACTICE TEAM STRUCTURE

PRIMARY LOCATION: MANCHESTER, UK DATA & AI DELIVERY HUB

Dedicated delivery centre supporting:

- Data engineering and lakehouse architecture design
- Analytics platform implementation and dashboard development
- Generative AI implementation (Setu-AI, Einstein, SAP Business AI)
- MLOps and model governance
- Technical recruitment for data engineering and MLOps roles

CROSS-FUNCTIONAL ENGAGEMENT STRUCTURE

Data Engineers: Designing and implementing lakehouse architectures, data pipelines, and governance

Analytics Engineers: Building semantic models and analytics platforms

ML Engineers: Implementing generative AI solutions and optimising models

Data Scientists: Developing predictive models and analytics

Solutions Architects: Designing end-to-end data and AI solutions for clients

Project Managers: Coordinating delivery across multiple workstreams

CONTINUOUS ENABLEMENT

Data & AI practice uses same curriculum for internal training that it delivers to client teams

Regular training on new platforms, tools, and ML techniques

Certification and specialisation pathways (Salesforce, SAP, cloud data platforms)

Knowledge sharing and community of practice for data professionals